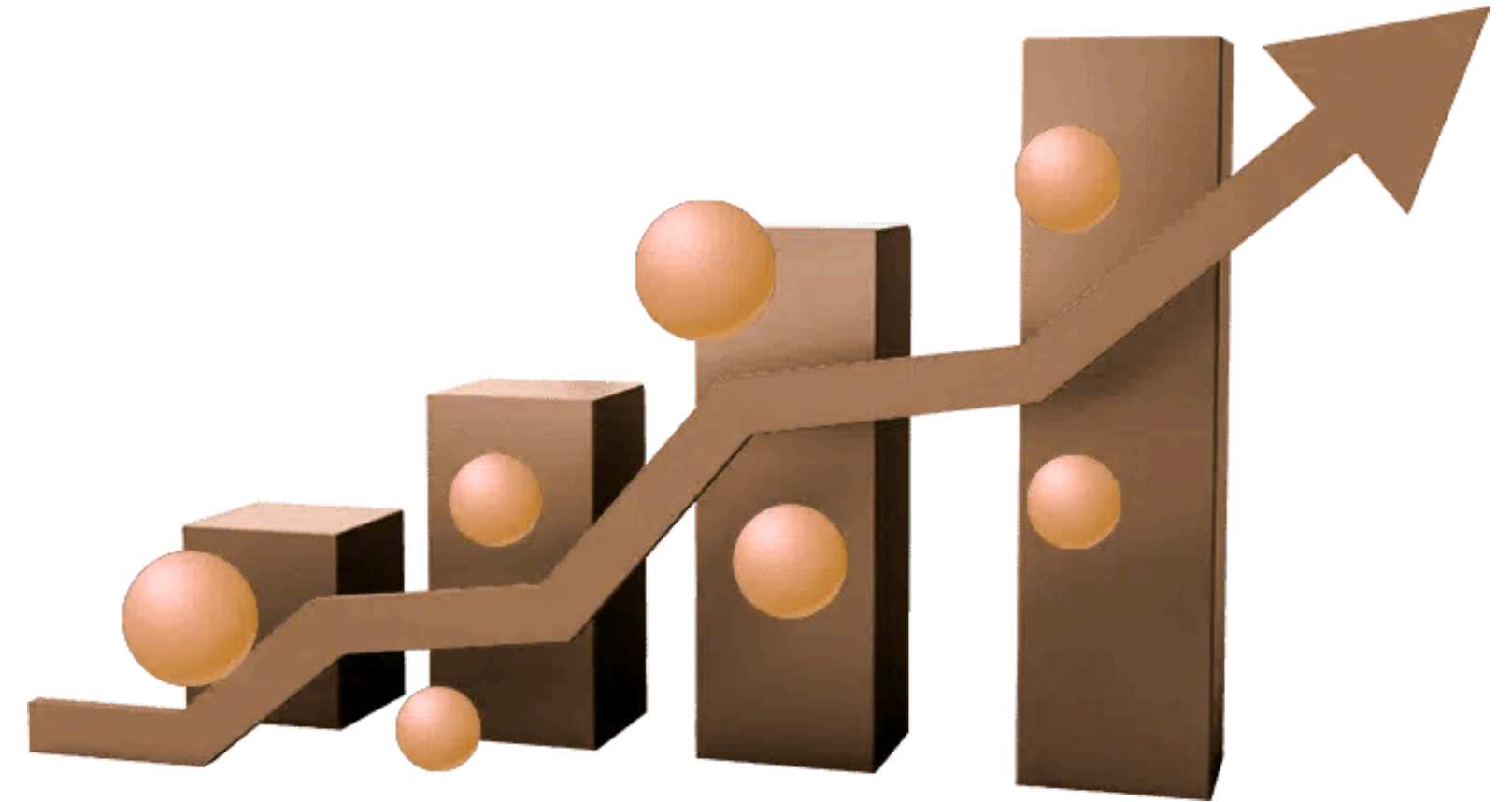


RELATIONSHIPS BETWEEN VARIABLES

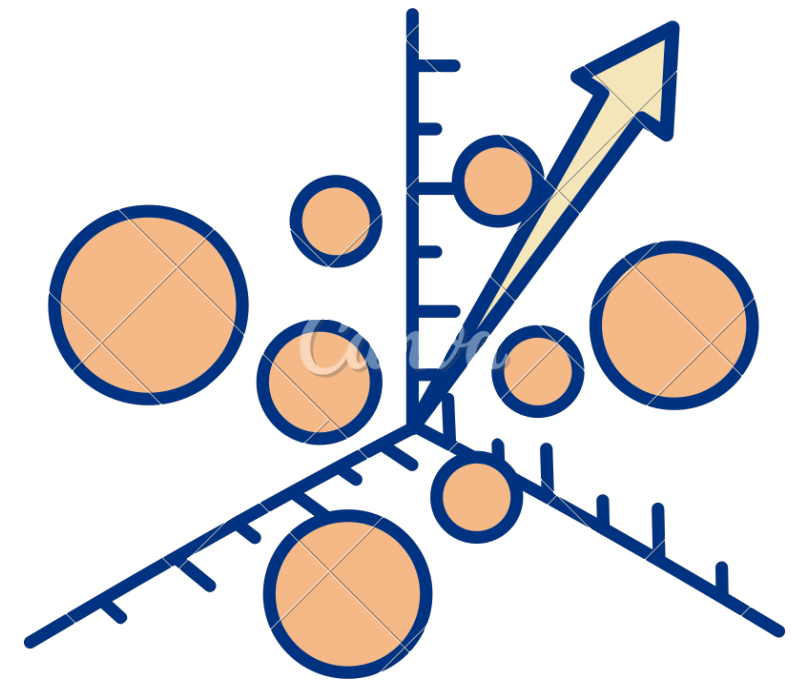


A comprehensive exploration of statistical methods for analyzing variable relationships

By / Fatema Taher Okasha

INTRO

In data analysis, understanding how variables relate to each other helps us find patterns, make predictions, and support decisions.



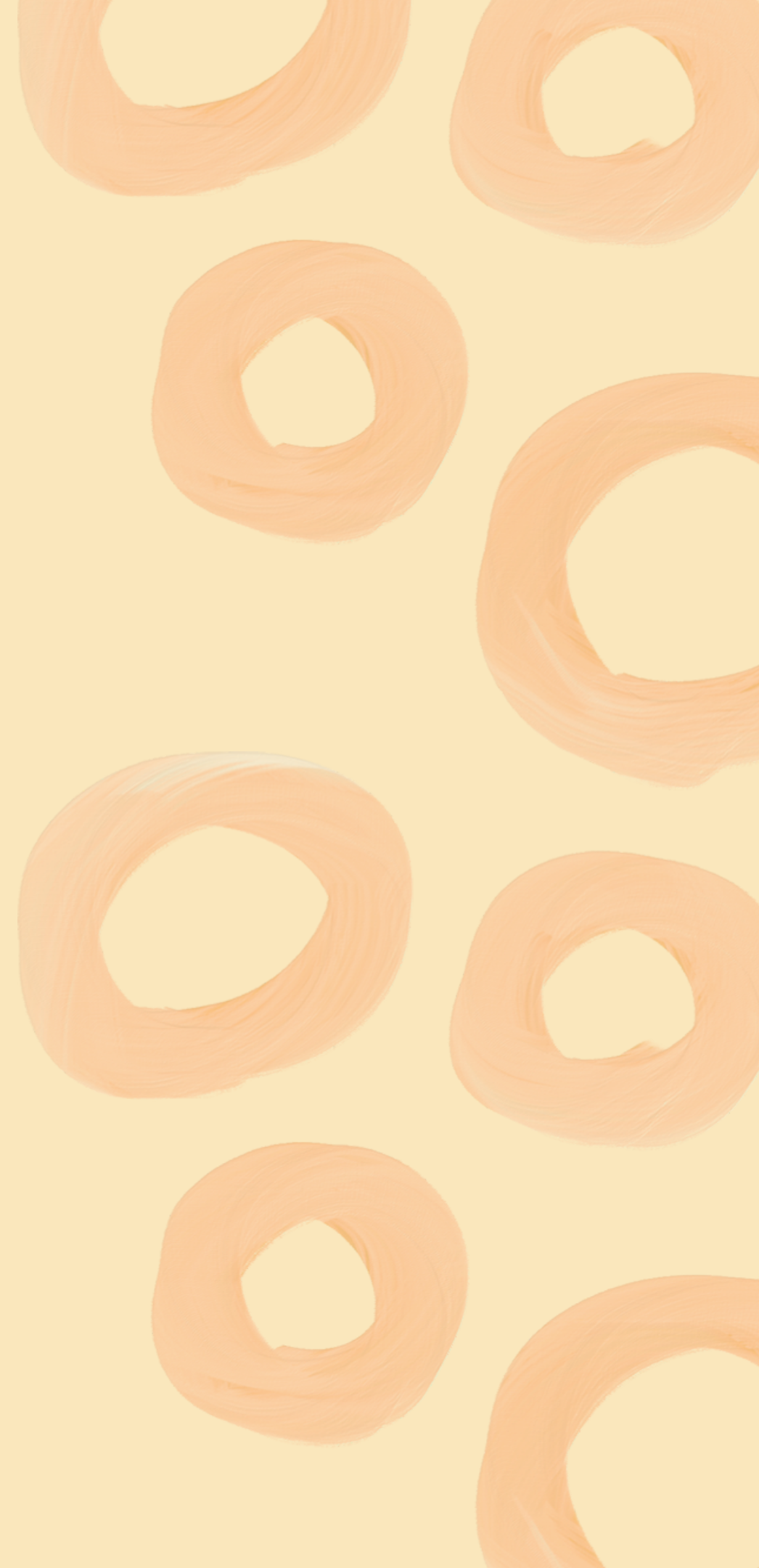
There are several statistical tools to measure these relationships, each serving a specific purpose

CONTENT

1. Correlation
2. Covariance
3. Regression
4. Mutual Info
5. Granger Causality
6. hypothesis testing



COVARIANCE



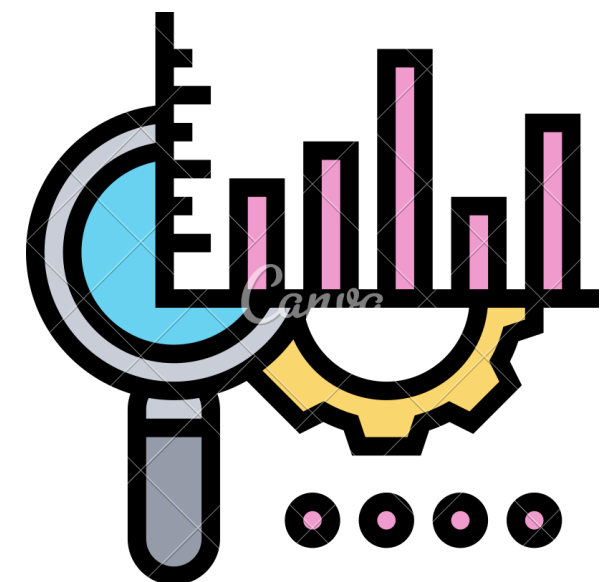
What is Covariance

Covariance measures the joint variability of two random variables. It indicates **whether variables tend to move together** (positive covariance) **or in opposite directions** (negative covariance).

$$COV(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

When to Use

- Initial exploration of variable relationships
- When direction of relationship is more important than strength
- As building block for correlation and regression



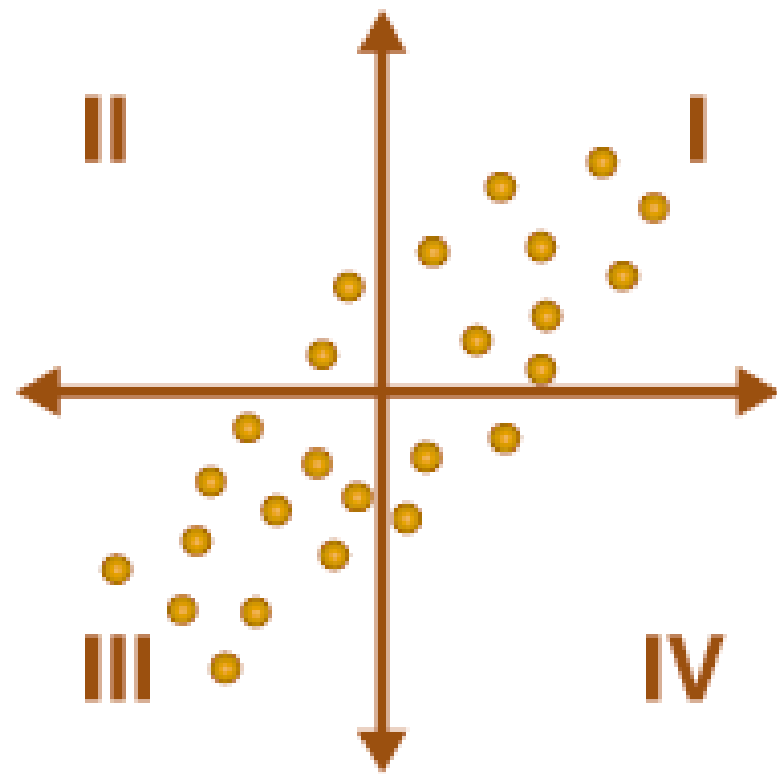
Key Properties

- **Directional:** Shows only direction of relationship (positive/negative)
- **Scale-dependent:** Value changes with units of measurement
- **Unbounded:** Ranges from $-\infty$ to $+\infty$
- **Symmetric:** $\text{Cov}(X,Y) = \text{Cov}(Y,X)$

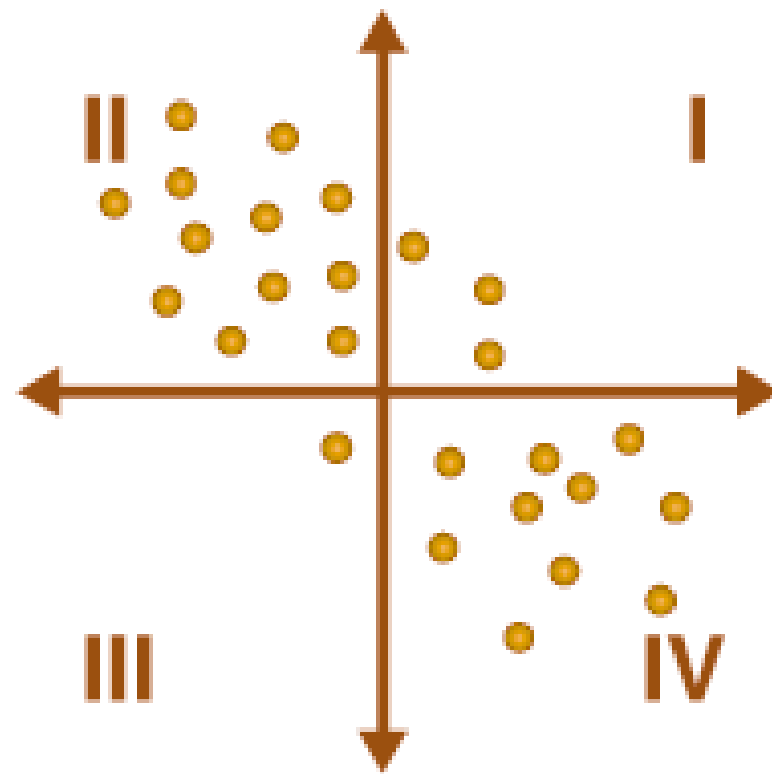
Example Code

```
import numpy as np

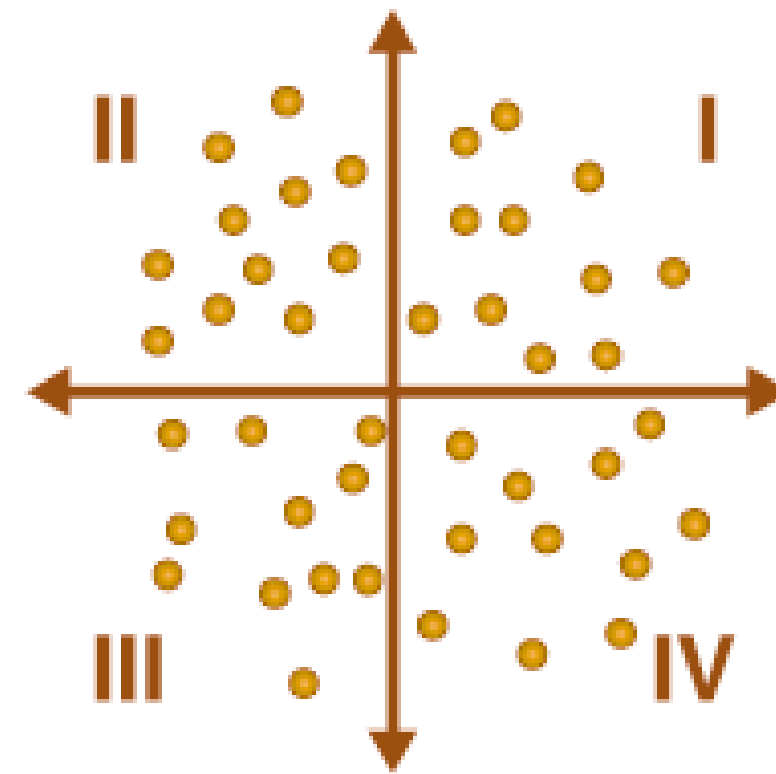
# Calculate covariance
x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 6, 8, 10])
cov_matrix = np.cov(x, y)
cov_xy = cov_matrix[0, 1] # Covariance between x and y
```



(A) Positive relationship



(b) Negative relationship



(c) No relationship



CORRELATION

What is Correlation

quantifies both the **strength and direction of the linear relationship** between two variables.

It standardizes covariance to a unitless measure.

Key Properties

- **Standardized:** Ranges from -1 to +1
- **Unitless:** Independent of measurement scale
- **Strength interpretation**
 1. $|r| < 0.3$: Weak correlation
 2. $0.3 \leq |r| < 0.7$: Moderate correlation
 3. $|r| \geq 0.7$: Strong correlation



Types of Correlation

Pearson Correlation:

- Measures linear relationships
- Assumes normally distributed data
- Sensitive to outliers

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Spearman's Rank Correlation:

- Measures monotonic relationships (linear or nonlinear)
- Based on ranked values

$$r_s = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

Kendall's Tau:

- Measures ordinal association
- Based on concordant and discordant pairs

$$\tau = \frac{C - D}{C + D}$$

When to Use

- Comparing strength of relationships across different variable pairs
- When standardized measure is needed
- Pearson: For linear relationships with normally distributed data
- Spearman: For monotonic relationships or ordinal data
- Kendall: For small sample sizes or many tied ranks

```
from scipy import stats
```

```
# Pearson correlation
```

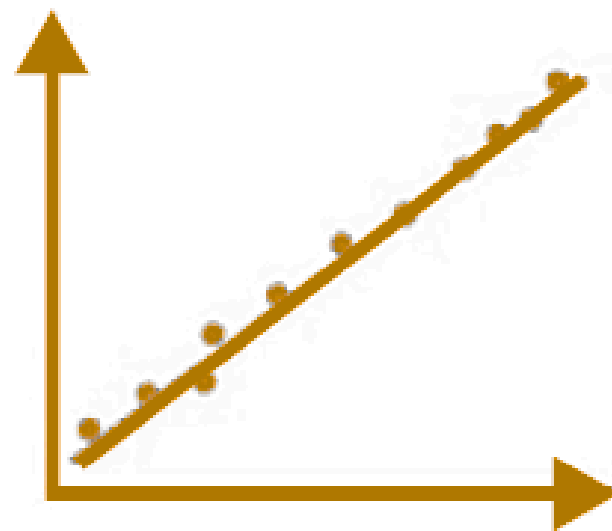
```
pearson_corr, _ = stats.pearsonr(x, y)
```

```
# Spearman correlation
```

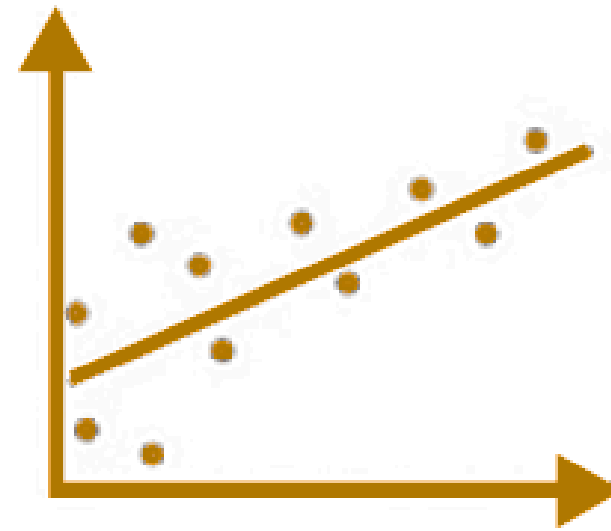
```
spearman_corr, _ = stats.spearmanr(x, y)
```

```
# Kendall's tau
```

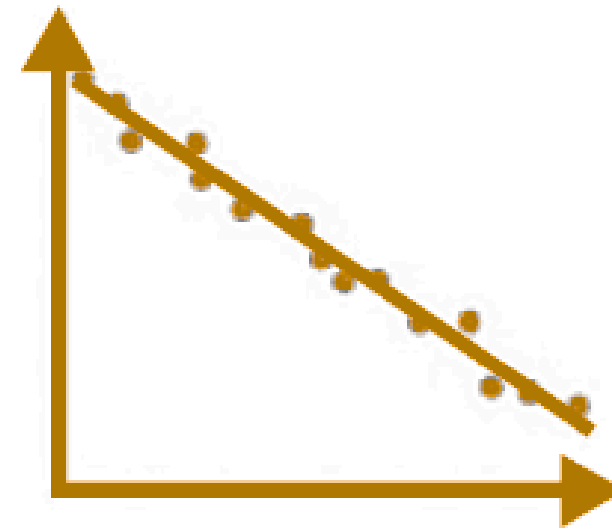
```
kendall_corr, _ = stats.kendalltau(x, y)
```



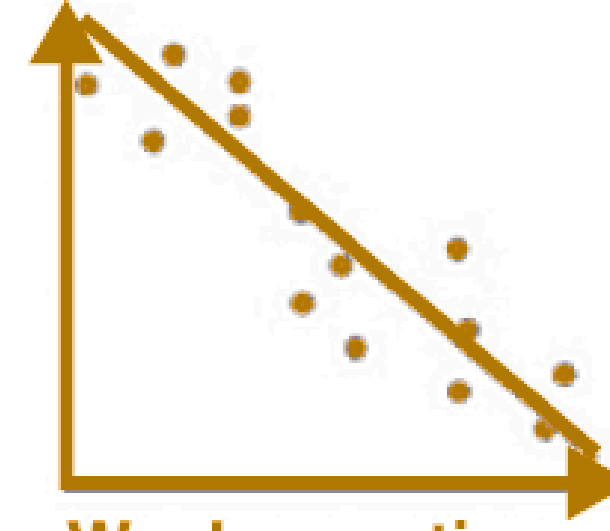
Strong positive correlation



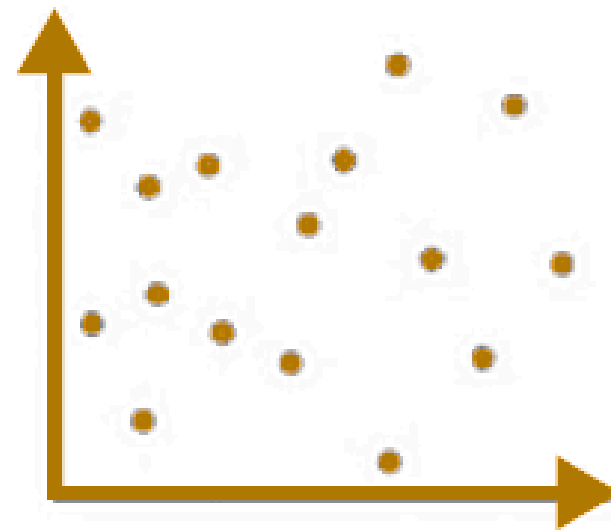
Weak positive correlation



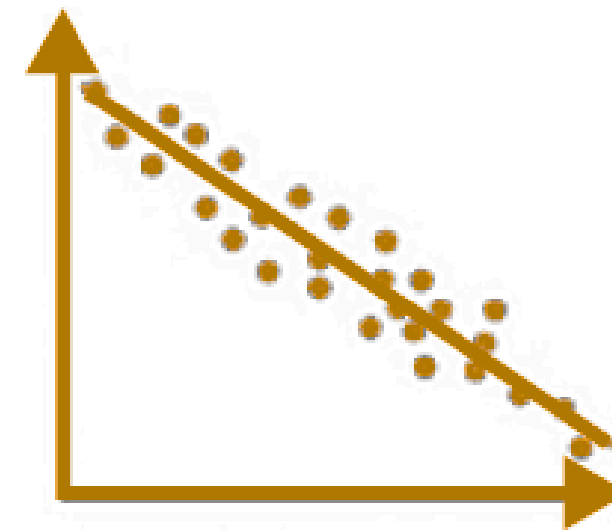
Strong negative correlation



Weak negative correlation



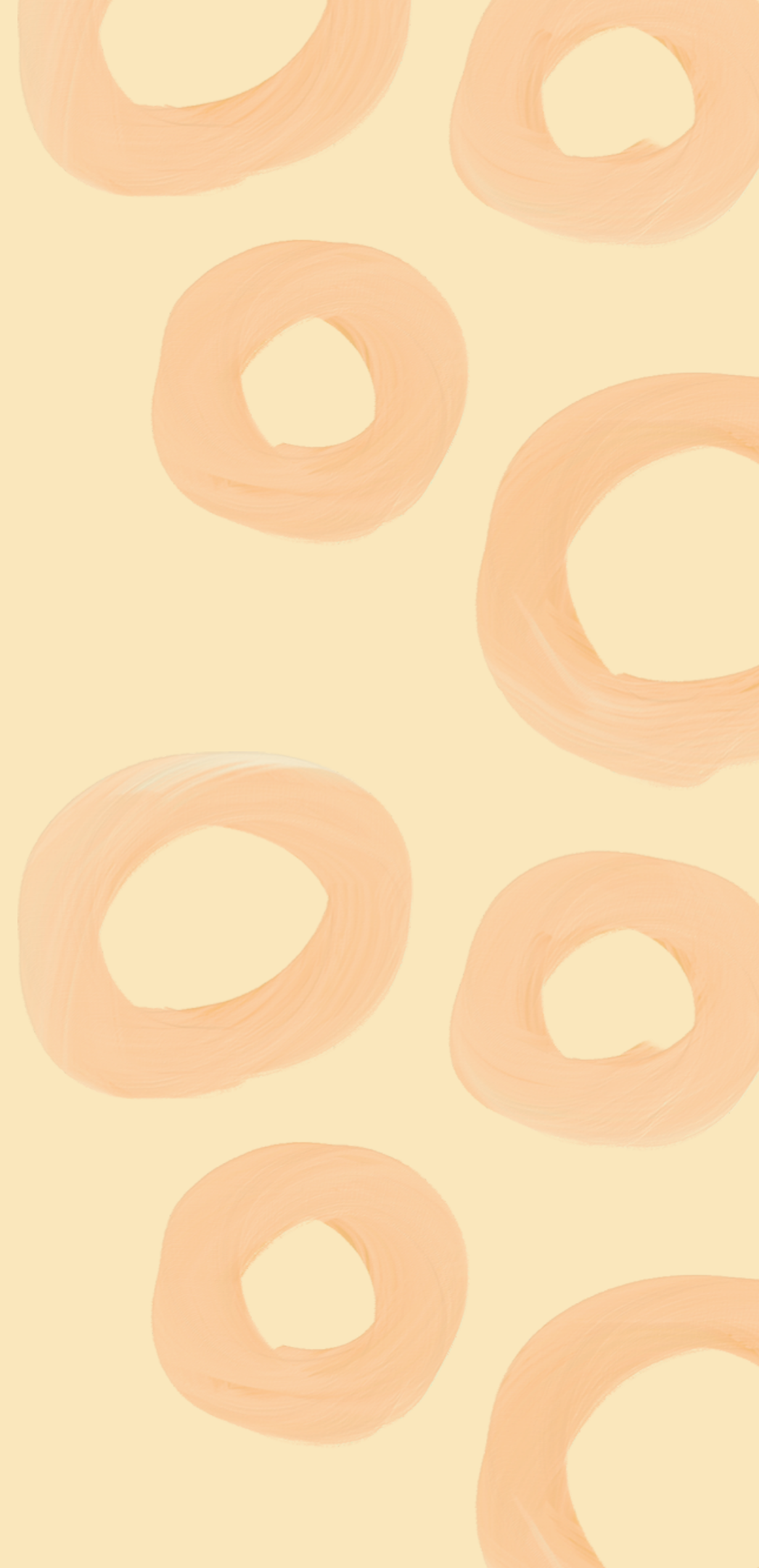
No correlation



Moderate negative correlation



REGRESSION ANALYSIS

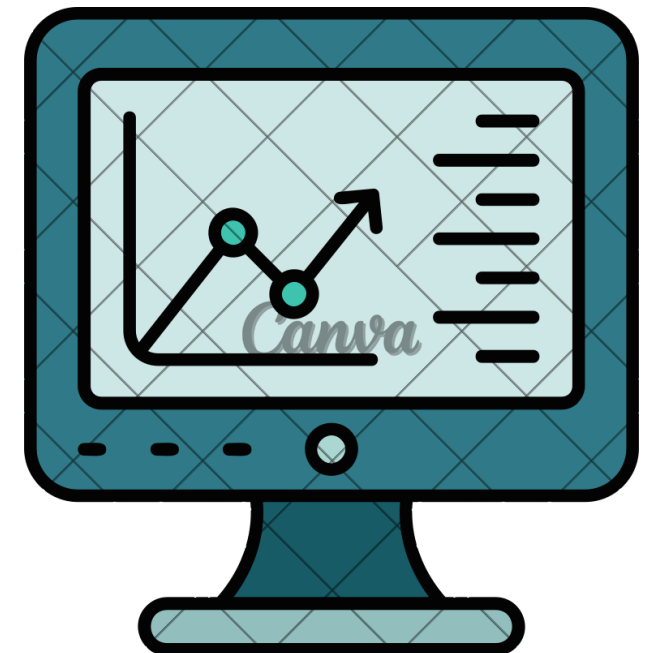


What is Regression analysis

Regression analysis models the **relationship between a dependent variable (Y) and one or more independent variables (X)**. It estimates the conditional expectation of Y given X.

When to Use

- Predicting values of dependent variable
- Understanding relationships between variables
- Testing hypotheses about variable relationships
- Controlling for confounding variables



Types of Regression

Simple Linear Regression:

- One independent variable

$$Y_i = \beta_0 + \beta_1 X_i$$

Multiple Linear Regression:

- Multiple independent variables

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$$

Polynomial Regression:

- Models nonlinear relationships

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + \dots + b_n x_1^n$$

Logistic Regression:

- For binary dependent variables

$$y = \frac{e^{(b_0 + b_1 X)}}{1 + e^{(b_0 + b_1 X)}}$$

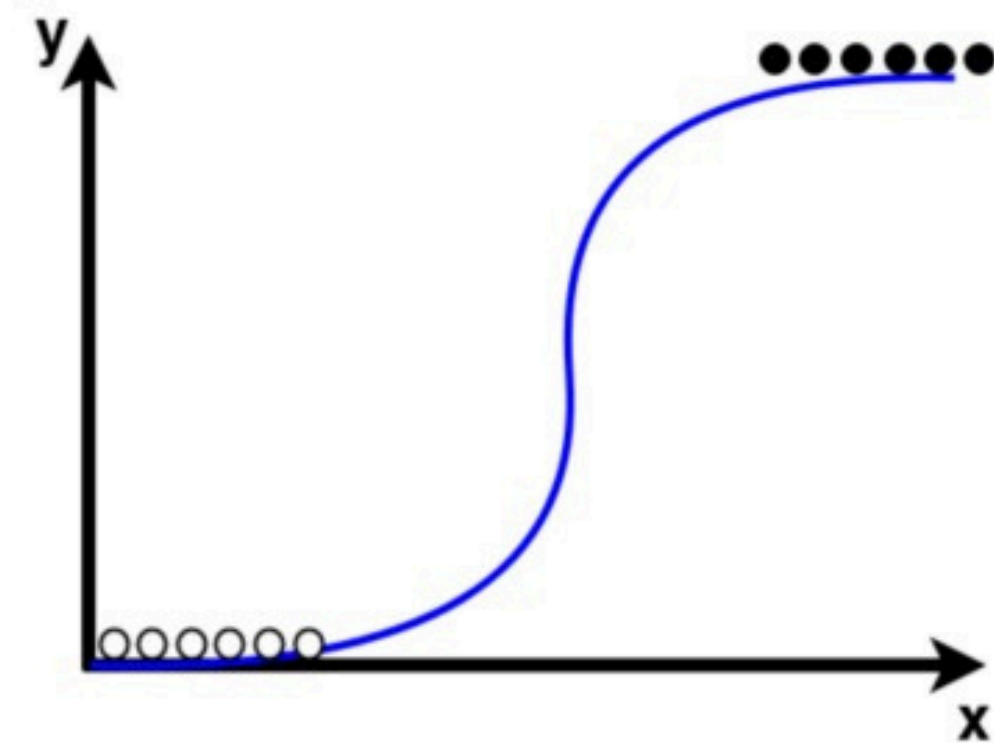
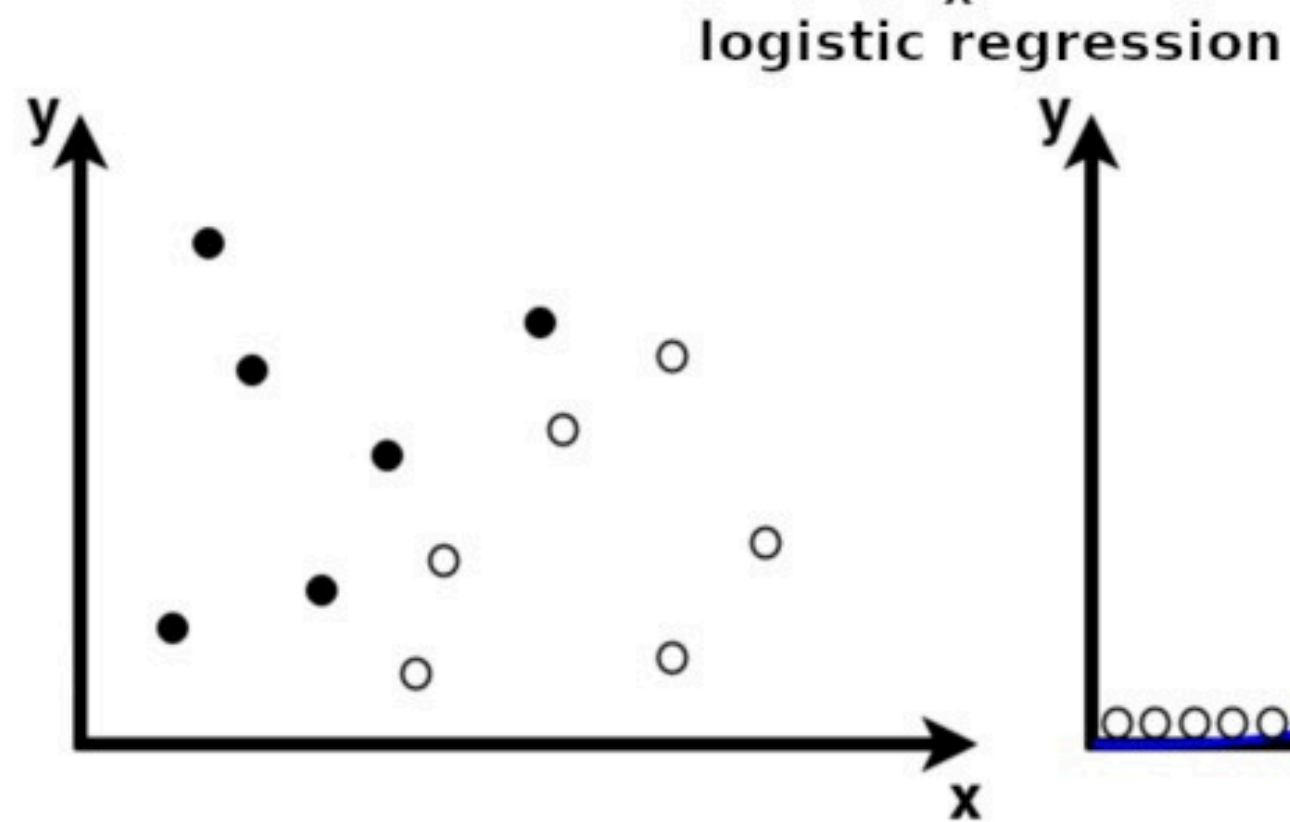
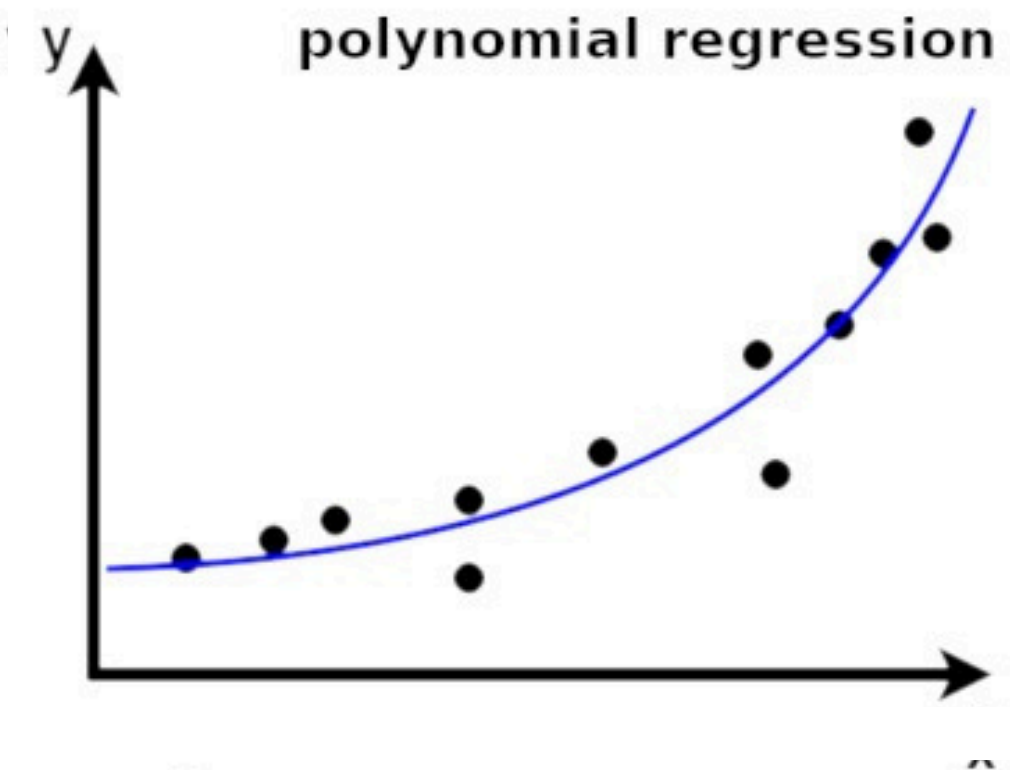
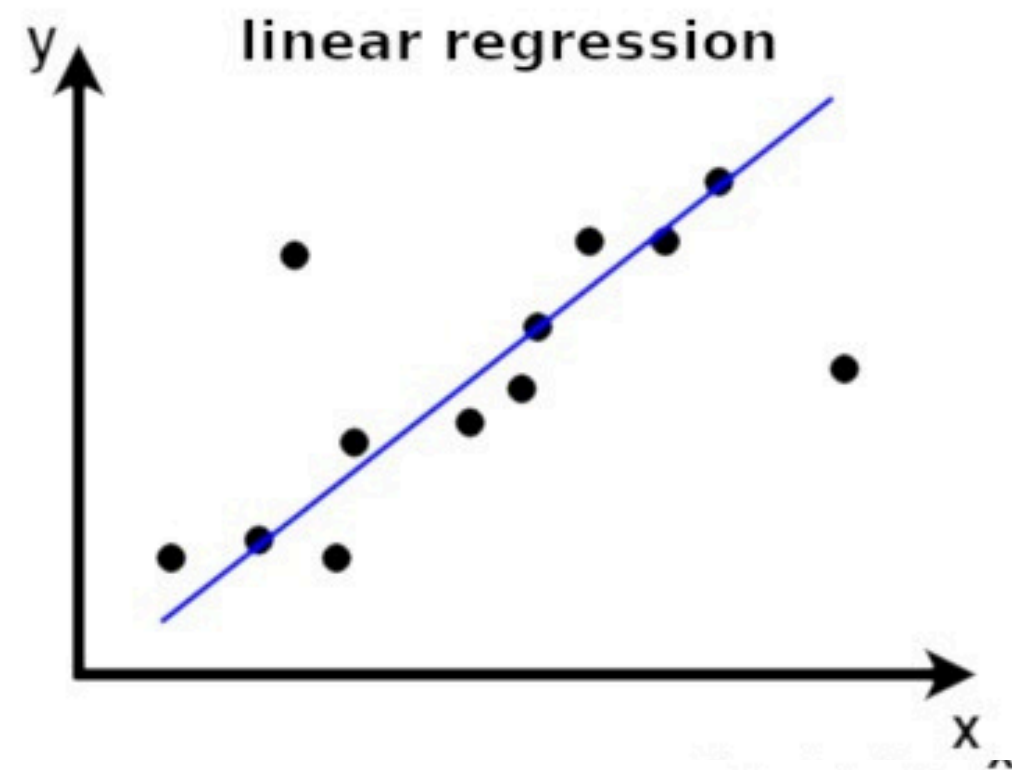
Example Code

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

# Simple linear regression
model = LinearRegression()
model.fit(X.reshape(-1, 1), y) # X needs to be 2D array

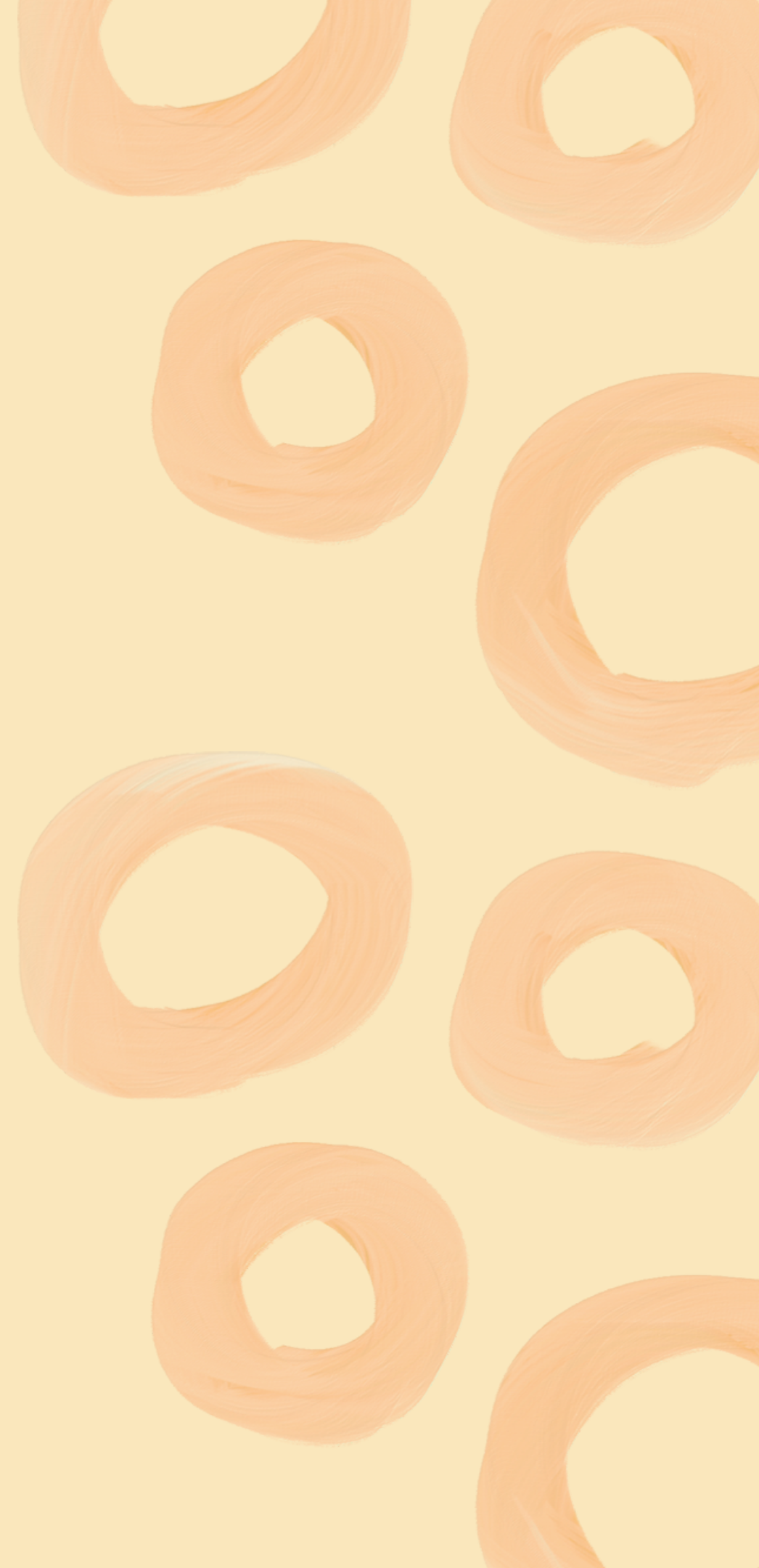
# Multiple linear regression
model.fit(X_multiple, y) # X_multiple has multiple
features

# Polynomial regression
poly_features = PolynomialFeatures(degree=2)
X_poly = poly_features.fit_transform(X)
model.fit(X_poly, y)
```



MUTUAL INFORMATION

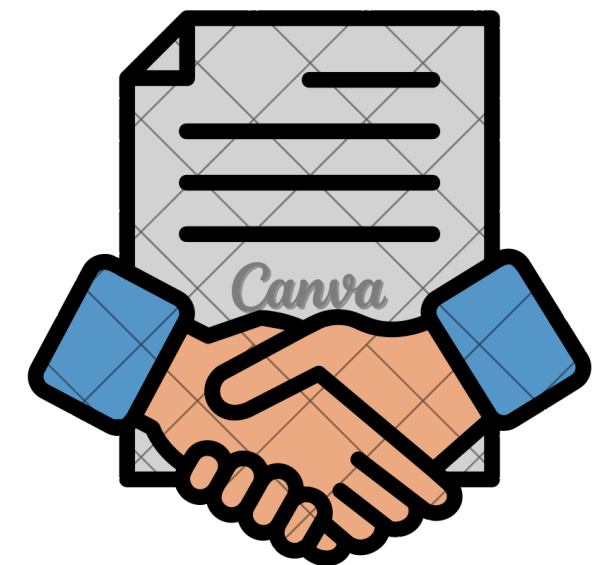


What is Mutual Information

Mutual Information (MI) measures the **amount of information obtained about one random variable** by observing another random variable. It quantifies the reduction in uncertainty about one variable given knowledge of another

When to Use

- Detecting nonlinear relationships
- Feature selection in machine learning
- Comparing dependencies between different variable pairs
- When relationship type is unknown
- With categorical or continuous variables



Formula

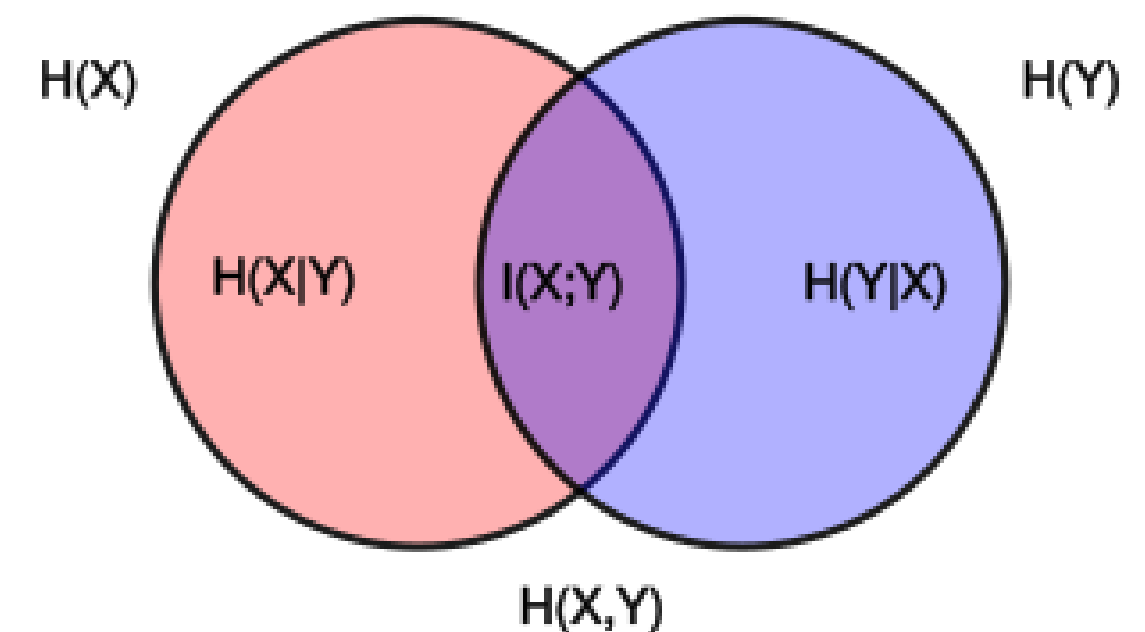
$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

- **(p(x,y))** is the joint probability distribution of (X) and (Y).
- **(p(x)) and (p(y))** are the marginal probability distributions of (X) and (Y), respectively

- It can also be expressed as the difference in entropy:

$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X).$$



Interpretation

- **$I(X;Y) = 0$** : Variables are independent
- **Higher values** indicate stronger dependence
- **No upper bound** (unlike correlation)

Key Properties

- **Non-negative**: $I(X;Y) \geq 0$
- **Symmetric**: $I(X;Y) = I(Y;X)$
- **Measures dependence**: Zero iff X and Y are independent
- **Captures all relationships**: Linear and nonlinear
- **Unitless**: Measured in bits, nats, or bans depending on log base

Example Code

```
from sklearn.feature_selection import mutual_info_regression
```

```
# For continuous variables
```

```
mi = mutual_info_regression(X.reshape(-1, 1), y)[0]
```

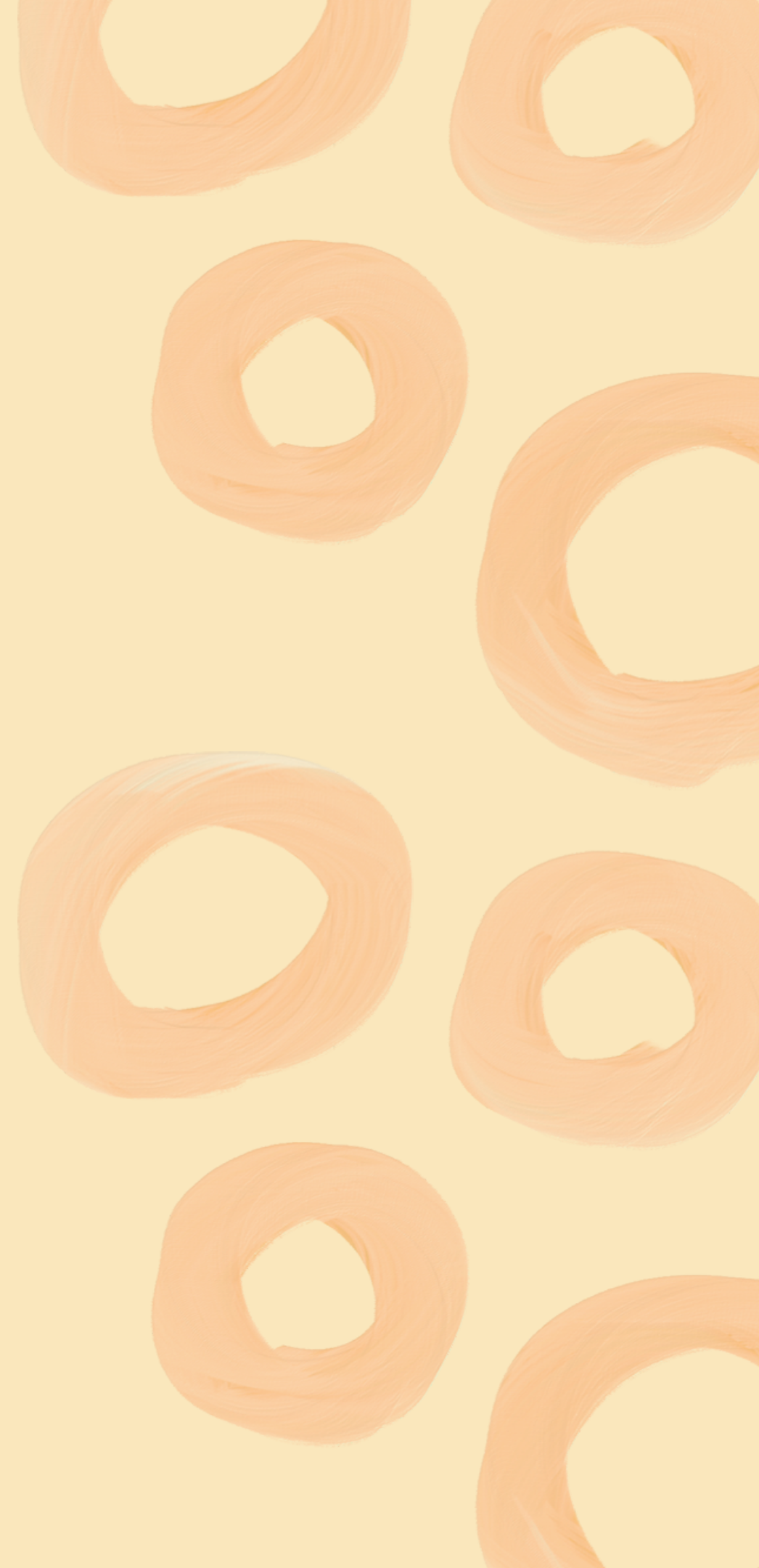
```
# For discrete variables
```

```
from sklearn.metrics import mutual_info_score
```

```
mi = mutual_info_score(x_discrete, y_discrete)
```



GRANGER CAUSALITY

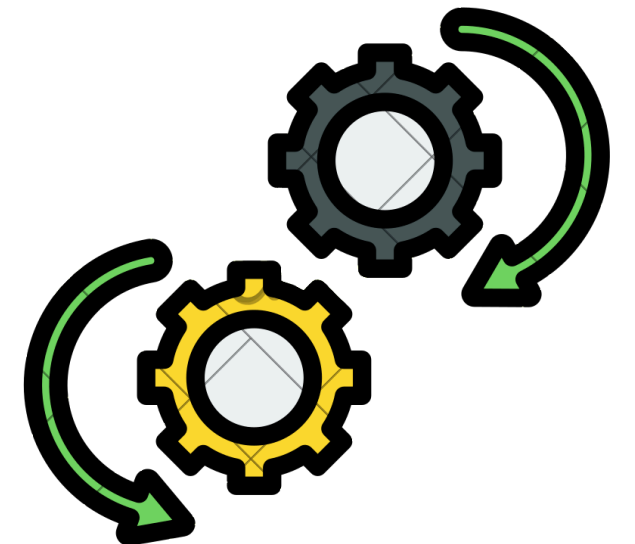


What is Granger Causality

Granger Causality tests whether **past values of one time series help predict future values** of another time series. It's a statistical test of predictive causality, not true causality.

Concept

Variable X "Granger-causes" Y if past values of X contain information that helps predict Y above and beyond the information contained in past values of Y alone.



Formula

Restricted model: Uses only past values of Y

$$Y_t = \alpha_2 + \sum_{i=1}^p \beta_2 Y_{t-i} + \epsilon_{2t}$$

UnRestricted model: Uses past values of X and Y

$$Y_t = \alpha_1 + \sum_{i=1}^p \beta_1 Y_{t-i} + \sum_{j=1}^q \gamma_1 X_{t-j} + \epsilon_{1t}$$

Where:

- RSS_R: Residual sum of squares from restricted model
- RSS_U: Residual sum of squares from unrestricted model
- p: Number of lags
- T: Number of observations

When to Use

- Analyzing time series data
- Testing for lead-lag relationships
- Econometric modeling
- Neuroscience (brain connectivity)
- Financial market analysis

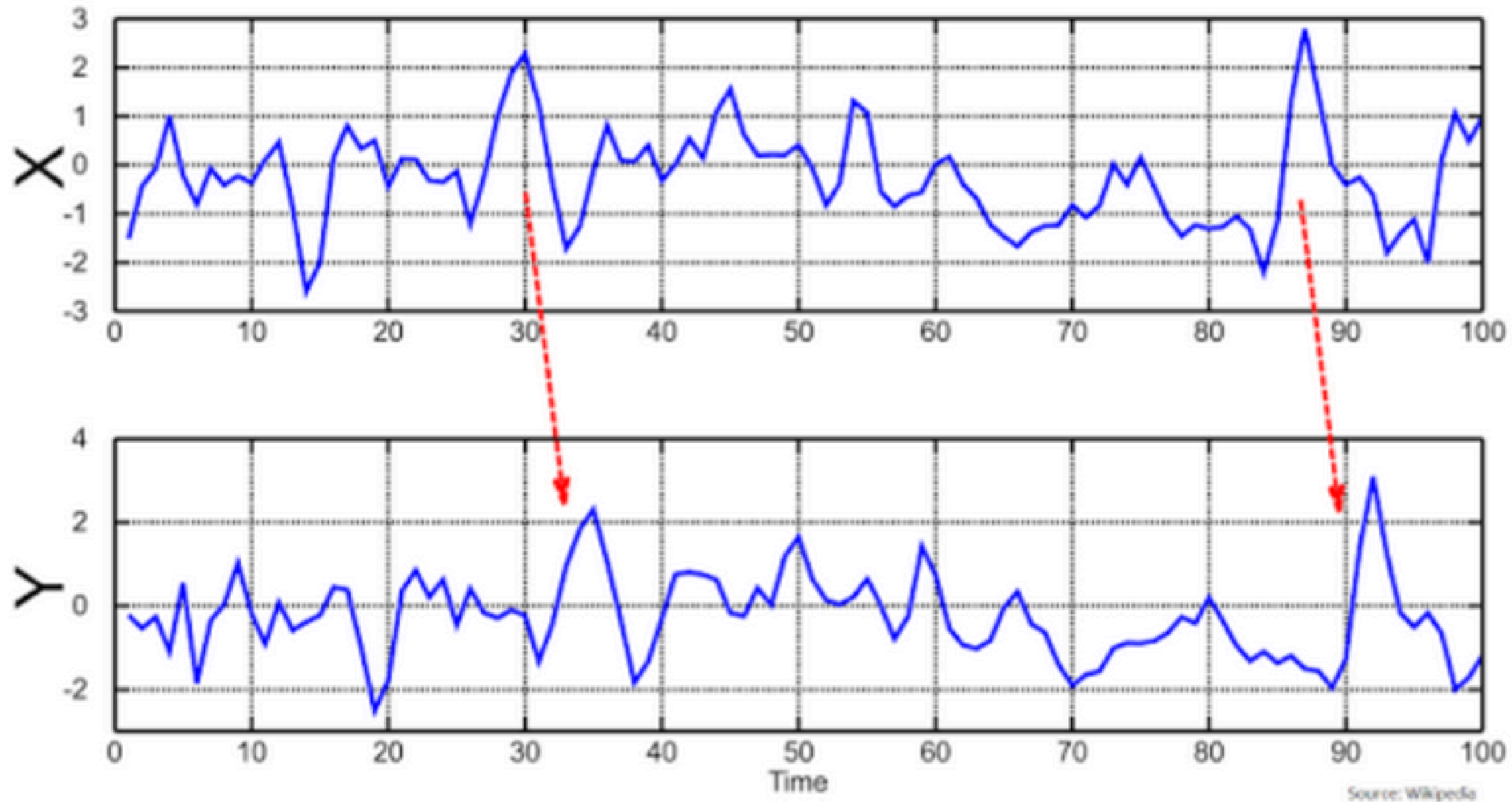
Key Properties

- **Temporal:** Requires time-ordered data
- **Predictive:** Measures predictive ability, not true causality
- **Directional:** Tests $X \rightarrow Y$ separately from $Y \rightarrow X$
- **Lag-dependent:** Results depend on chosen lag length

Example Code

```
from statsmodels.tsa.stattools import grangercausalitytests

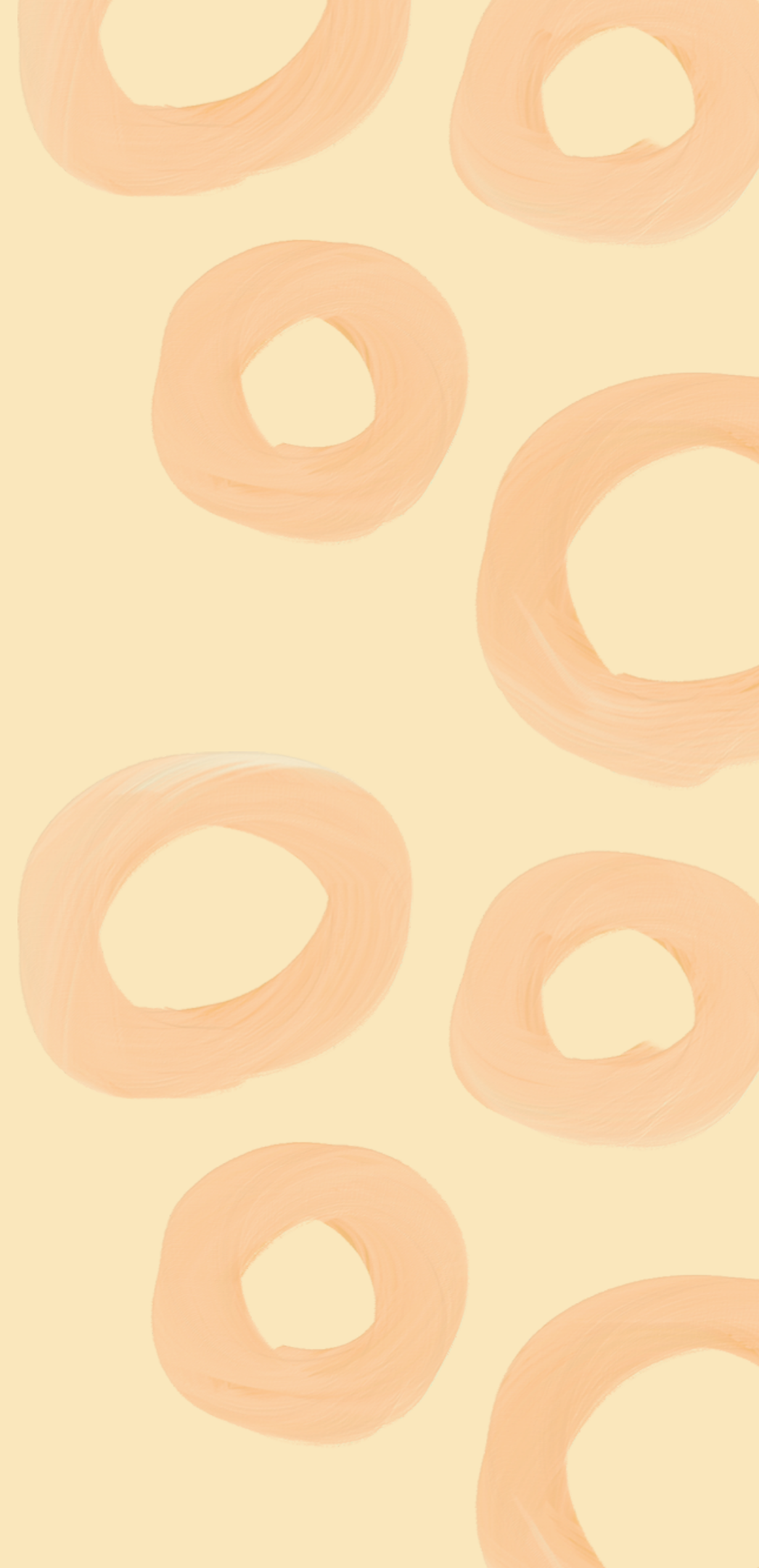
# Test if X Granger-causes Y
gc_result = grangercausalitytests(data[['Y', 'X']], maxlag=3)
```



This does not mean that X physically causes Y in a real-world causal sense — only that X contains useful information about the future behavior of Y beyond Y's own past values.



HYPOTHESIS TESTING



What is Hypothesis Testing

Hypothesis Testing is a statistical framework used to determine whether the patterns or relationships we see in sample data are real (significant) or just happened by chance.

When to Use

- Whether two or more groups are different
- Whether two variables are related
- Whether a pattern found in sample data is likely true for the entire population.



Types of Hypothesis Testing

1. Parametric Tests
2. Non-Parametric Tests
3. Regression-based Tests
4. Proportion Tests
5. Categorical Data Tests

Parametric Tests

used when the data is normally distributed and has no major outliers
mainly compare means or measure linear relationships between numerical variables

t-test

Compares the means of two groups

ANOVA (Analysis of Variance)

Used to compare the means of three or more groups to see if at least one group differs significantly.

z-test

used when the population standard deviation is known, and the sample size is large ($n > 30$).

Pearson Correlation

Measures the strength and direction of a linear relationship between two numerical variables. The coefficient ranges from -1 to +1.

Non-Parametric Tests

These are used when data does not follow a normal distribution, contains outliers, or involves ordinal data (ranked values). They are more flexible but less powerful statistically



Spearman Correlation:

A rank-based version of Pearson's correlation, measuring monotonic (not necessarily linear) relationships between two variables.



Kruskal-Wallis Test: A non-parametric version of ANOVA, comparing ranks among three or more groups.



Mann-Whitney U Test

A non-parametric alternative to the independent t-test, comparing ranks of two groups.

Regression-based Tests

Regression tests explore how one or more independent variables (X) affect a dependent variable (Y). They are used for prediction and causal inference.



Linear Regression

Examines the linear relationship between one independent variable and one dependent variable.



Multiple Regression

Extends linear regression to include multiple independent variables, analyzing how each contributes to predicting Y.

Proportion Tests

These tests are applied when comparing ratios or percentages instead of means. They're common in medical studies, surveys, or marketing analysis



One-Proportion z-test:

Tests whether a single proportion equals a specific value (e.g., is the success rate really 70%).



Two-Proportion z-test:

Compares proportions from two groups (e.g., percentage of smokers among men vs. women).

Categorical Data Tests

These tests are used for categorical or qualitative variables, where values represent categories like “Yes/No” or “Male/Female” instead of numbers



Chi-square (χ^2) Test

Compares observed frequencies with expected frequencies in contingency tables. It's used to test whether two categorical variables are related.



Fisher's Exact Test:

Used instead of Chi-square when sample sizes are small. It calculates the exact probability of observing the data pattern

Example Code

```
from scipy import stats
import pandas as pd

stockA = [0.05, 0.03, 0.06, 0.02, 0.04]
stockB = [0.01, 0.02, 0.00, 0.03, 0.01]

t_stat, p_value = stats.ttest_ind(stockA, stockB)

if p_value < 0.05:
    print("Significant difference between the two stocks.")
else:
    print("No significant difference (probably random).")
```


THANK YOU!



FatemaTaher



fatematahersayed@gmail.com



fatema-taher-356455258

Any
question?

